

Stage 5 — Practice, breadth and theory

Where the Shortcut Sits: progress report 5

In one sentence. On unaltered test sets masking harms exactly the cases the location law says it should — shortcut-conflicting images with an in-ROI artifact — and on the patient-disjoint thyroid split it lowers sensitivity at every validation-fixed operating point; the crossover replicates in 17 of 18 frozen encoder-cohort pairs, at three model sizes, in 3 of 5 fine-tuned networks, in the zero-shot scores of vision-language models, and on new patients (ISIC 2020, +0.250 [+0.174, +0.327]); a two-parameter linear-Gaussian model predicts held-out reversed AUROC with mean error 0.052, less than half the error of the best heuristic, and explains why the law holds.

Notation

$Y \in \{0, 1\}$	diagnosis: 1 = melanoma, malignant nodule, tumour with a solid component, erosion.
$A \in \{0, 1\}$	artifact present: a real artifact: hair, sonographer caliper, capsule debris.
ROI	region of interest: lesion, nodule, tumour, union of expert lesion boxes; drawn in green ; artifacts in red .
r	overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $. Trap A: $r \geq 0.5$ (artifact in the ROI); Trap B: $r < 0.1$ (outside).
Environments	training and correlated test $P(A=1 \mid Y=1) = 0.9$, $P(A=1 \mid Y=0) = 0.1$; reversed test 0.1/0.9; clean test: artifact independent of Y (real artifacts: half of each class) or absent (synthetic artifacts).
ERM, mask	linear head on the frozen features of the original image / of the ROI-masked image.
Crossover	$[\text{mask} - \text{ERM}]_{\text{Trap B}} - [\text{mask} - \text{ERM}]_{\text{Trap A}}$ of reversed-test AUROC.
CI	95% crossed seed×image bootstrap interval (Stage 1); thresholds are re-estimated in every replicate for operating points.
$ \Delta p $	same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted into the same image.
Hard pairs	positive-negative pairs whose artifact status conflicts with the test set's own artifact-label association (the cases a shortcut gets wrong).
OP1-OP5	decision thresholds fixed on validation data: maximal balanced accuracy; specificity ≥ 0.80 , ≥ 0.90 ; sensitivity ≥ 0.80 , ≥ 0.90 .

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1 Where this stage starts

Stages 1–4 established the location law in constructed environments — a strong artifact–label association in training and its reversal at test — showed its mechanism (retention with amplified reliance), and showed that it is not explained by which images carry the artifact. A clinician will ask two questions that this cannot answer: does it matter on data as collected, without an imposed association, and at the operating point a clinic would use? A methodologist will ask two more: is the result a property of one encoder and one training regime, and can it be predicted rather than only observed? This stage answers all four.

2 Questions

- Q5a** On natural test sets, does masking harm the shortcut-conflicting cases, and does that change clinical sensitivity at fixed operating points?
- Q5b** How broadly does the crossover replicate: encoder families and sizes, end-to-end fine-tuning, untrained (zero-shot) models, new patients?
- Q5c** Can a simple model predict a model’s reversed-test AUROC from what is observable without the reversed test — and so explain when masking helps?

Design decisions *Hard pairs.* On a natural test set the artifact–label association is weak, so the overall AUROC barely moves. Positive–negative pairs are therefore split by whether their artifact status conflicts with the test set’s own association (hard) or follows it (easy); the law predicts harm on hard pairs, where the retained in-ROI artifact points the wrong way. *Operating points fixed on validation data* (never on test) at five clinically motivated targets, with thresholds re-estimated in every bootstrap replicate. *A theory judged by held-out error, not by fit:* the model is fitted to each cell’s clean and correlated AUROC and must predict the reversed AUROC it never saw, against two parameter-free heuristics. Rejected: judging the theory by agreement with simulations drawn from its own assumptions (circular), and by the correlation of predicted and observed crossovers (a heuristic reaches almost the same correlation).

3 Methods

Natural test sets Models are trained on the natural training distribution (no imposed association). Thyroid: the official patient-disjoint TNCD split (Gong et al., 2022) (614 test images; in-ROI calipers on 37.6% of benign and 24.1% of malignant training images). Dermoscopy: each source (BCN, HAM, MSK) held out in turn (in-lesion hair on 31% of melanomas and 20% of benign lesions). Capsule: held-out frame-block groups. 5 seeds of a group-safe train/validation split. A = in-ROI artifact. Arms ERM, masking and group balancing.

Clinical metrics and operating points AUROC, AUPRC, Brier score, expected calibration error; sensitivity and specificity at OP1–OP5, overall and among shortcut-conflicting positives (for thyroid: malignant nodules with a caliper on the nodule).

Replication Frozen DINOv2 ViT-S/B/L (21/86/304 M parameters), MedSigLIP-448, ConvNeXt-384 and DermLIP with linear heads; ResNet-50 and ViT-S/16 fine-tuned end to end (ImageNet initialisation, 224 px, 8 epochs, AdamW, one-cycle learning rate 10^{-4} or $3 \cdot 10^{-5}$, head $\times 10$, thresholds on clean validation; the five folds of seed 42 are the bootstrap clusters, 15 clusters for the ovary); zero-shot vision–language scores (cosine to a positive minus a negative class prompt) of MedSigLIP and DermLIP.

External validation ISIC 2020 (Rotemberg et al., 2021) was run only if a gate registered before any external label was read held: direct download without an agreement, a research licence, patient identifiers, masks producible by frozen models, and at least 50 images in every cell. Lesion masks: the existing U-Net (ISIC 2018 Task 1 only). Hair masks: a segmenter trained only on the ISIC 2019 hair masks and frozen before any ISIC 2020 image was scored. Protocol of the ISIC 2019 traps unchanged; groups = patient identifier.

Theory A frozen encoder maps an image to a disease block with signal-to-noise ratio S , an artifact block with signal-to-noise ratio A , and noise. Training uses $P(A=1 | Y=1) = p_1 = 0.9$ and $P(A=1 | Y=0) = p_0 = 0.1$. An LDA-optimal linear head (which logistic regression approximates) weights each block by its training signal; the artifact’s effective training signal is

$$\tilde{A} = \frac{(p_1 - p_0)^2 A}{1 + p_1(1 - p_1)A}, \quad \text{AUROC}_{\text{rev}} = \Phi \left(\frac{S - \tilde{A}}{\sqrt{2(S + \tilde{A})}} \right),$$

which rises with S and falls with $\tilde{A}$; for any test composition $q_y = P(A=1 | Y=y)$ the same model gives the AUROC in closed form (`docs/THEORY.md`). Its consequences: (1) out-of-ROI masking sets $\tilde{A} \rightarrow 0$ and helps unless it destroys most of the disease signal; (2) in-ROI masking keeps $\tilde{A}$ and changes S to S_m , so it *harms iff it removes disease-informative context* ($S_m < S$); (3) for the same S_m the crossover is positive for every $\tilde{A} > 0$; (4) group balancing sets the effective association to zero for the artifact and everything correlated with it, independently of location; (5) a shortcut carried by correlates that the clean and correlated environments do not reveal is invisible to the model. *Test*: for every cell (result $\times$ trap or overlap $\times$ arm) (S, A) is fitted to the observed clean and correlated AUROC and the reversed AUROC is predicted; heuristics “reversed = clean” and “symmetric reversal” (reversed = clean – (correlated – clean)).

4 Experiments

Table 1: Experiments of this stage and their registration documents (docs/PREREGISTRATION_<name>.md). Commit times are set by the committing machine and are not independent proof of order.

Experiment	Status	Registration	First commit (local time)	Support criterion
Zero-shot vision-language scores, Z1	partly supported registered (descriptive); crossover test post hoc	TEXT_PROMPT	7877e8c, 12:21	2026-09-27 correlated — reversed gap reported
Natural test sets, N3 (hard pairs)	supported registered	NATURAL	74bd243, 09:39	2026-09-27 mask — ERM < 0 on hard pairs
Operating points S1–S3	supported registered	REVIEW2 (R3)	29cd0da, 18:56	2026-09-27 “masking lowers sensitivity” only if S1 and ≥ 3 of OP2–OP5
Thyroid subgroup, $n=78$ (S3)	partly supported registered test in a post hoc subgroup	REVIEW2 (R3)	29cd0da, 18:56	2026-09-27 sensitivity loss in the subgroup, CI < 0
Crossed CIs for the subgroup (R6)	supported registered	REVIEW3 (R6)	85da735, 23:43	2026-09-27 report the crossed CI
Fine-tuned hair model and literature comparison (R4)	supported registered	REVIEW2 (R4)	29cd0da, 18:56	2026-09-27 crossover CI > 0; comparison descriptive
Encoder scale (S/B/L)	supported registered	SCALE	13303b0, 14:13	2026-09-27 crossover CI > 0 at every scale
Fine-tuned networks	supported registered	FINETUNE	7c8b466, 07:06	2026-09-27 crossover CI > 0 (secondary)
DermLIP hair traps	supported registered	ISIC2019-TRAPS	76e25d4, 13:14	2026-09-26 crossover CI > 0
Theory as held-out predictor, T1–T4	supported registered	THEORY_PREDICTION	4f78653, 13:46	2026-09-27 MAE below both heuristics; crossover signs
External validation, X1 (ISIC 2020)	supported registered	FINAL (A4)	c45d482, 00:35	2026-09-28 gate, then crossover CI > 0

5 Results

5.1 Natural test sets

Table 2. On all pairs, masking changes little. On shortcut-conflicting pairs it harms in thyroid (-0.095 [$-0.165, -0.022$]), BCN (-0.025 [$-0.043, -0.008$]) and MSK (-0.065 [$-0.111, -0.019$]), and helps in HAM ($+0.050$ [$+0.030, +0.070$]); in capsule the interval includes zero ($+0.021$ [$-0.003, +0.048$]). Where masking harms the hard pairs, it helps or is neutral on the easy ones (MSK $+0.100$ [$+0.056, +0.147$]) — the signature of a model that still uses an in-ROI artifact after masking: it gets the aligned cases right for the wrong reason and the conflicting cases wrong. The registered claim N3 is supported in three of five natural test sets. Group balancing improves on masking for the hard pairs of thyroid ($+0.136$ [$+0.063, +0.207$]), BCN ($+0.043$ [$+0.026, +0.059$]) and MSK ($+0.100$ [$+0.055, +0.144$]), but not on all pairs and not in capsule (-0.086 [$-0.110, -0.065$] on all pairs).

Table 2: Natural test sets: AUROC change from masking on all positive–negative pairs and on the pairs whose artifact status conflicts with (“hard”) or follows (“easy”) the test set’s own artifact–label association (DINOv2, crossed CIs).

Test set (no imposed correlation)	$P(A=1 Y=1) / P(A=1 Y=0)$	mask – ERM, all pairs	shortcut-conflicting pairs	shortcut-aligned pairs
Thyroid (official patient-disjoint split)	0.331 / 0.434	−0.004 [−0.048, +0.041]	−0.095 [−0.165, −0.022]	+0.042 [−0.011, +0.097]
Dermoscopy, BCN held out	0.356 / 0.239	−0.025 [−0.035, −0.016]	−0.025 [−0.043, −0.008]	−0.022 [−0.035, −0.008]
Dermoscopy, HAM held out	0.274 / 0.184	+0.031 [+0.017, +0.045]	+0.050 [+0.030, +0.070]	+0.004 [−0.019, +0.027]
Dermoscopy, MSK held out	0.109 / 0.074	+0.016 [−0.007, +0.038]	−0.065 [−0.111, −0.019]	+0.100 [+0.056, +0.147]
Capsule (held-out frames)	0.101 / 0.232	+0.077 [+0.059, +0.097]	+0.021 [−0.003, +0.048]	+0.071 [+0.040, +0.108]

5.2 Clinical consequences: thyroid operating points

On the official, patient-disjoint thyroid test split the AUROC of ERM and of the masked model is almost the same (0.735 and 0.731), but masking lowers sensitivity at every operating point (Table 3, Figure 1): at OP1 by -0.275 [−0.435, −0.138] (from 0.699 to 0.424) and at OP2–OP5 with intervals below zero in 4 of 4. By the decision rule registered before the operating points were computed (S1 and a loss at ≥ 3 of OP2–OP5), the thesis may say that masking lowers sensitivity on this test set. Specificity rises: masking moves the operating point, and the loss falls on the malignant nodules whose caliper lies on the nodule.

The subgroup of 78 malignant nodules **Registration status: a pre-registered test inside a post hoc subgroup — exploratory confirmation.** The subgroup (malignant nodules with an in-ROI caliper) was defined after the natural-distribution AUROC results were known; the operating-point test in it was registered before the operating points were computed. The same 78 nodules enter every seed. At OP2, sensitivity in this subgroup falls from 0.549 to 0.177, a change of -0.372 [−0.499, −0.236]. The effect is large and consistent across operating points; its confirmatory weight is limited by the post hoc subgroup definition.

Table 3: Thyroid, official patient-disjoint test split: effect of masking at five validation-fixed operating points (crossed bootstrap with thresholds re-estimated in every replicate). The subgroup is the malignant nodules with an in-ROI caliper (shortcut-conflicting).

Operating point (fixed on validation)	Sensitivity ERM → mask	Δ sensitivity	Δ specificity	Sensitivity, conflicting malignant (ERM → mask)	Δ in that subgroup
OP1 max. balanced accuracy	0.699 → 0.424	−0.275 [−0.435, −0.138]	+0.202 [+0.115, +0.330]	0.703 → 0.223	−0.479 [−0.640, −0.313]
OP2 specificity ≥ 0.80	0.557 → 0.355	−0.202 [−0.300, −0.100]	+0.117 [+0.059, +0.172]	0.549 → 0.177	−0.372 [−0.499, −0.236]
OP3 specificity ≥ 0.90	0.371 → 0.219	−0.152 [−0.253, −0.043]	+0.075 [+0.029, +0.122]	0.336 → 0.100	−0.236 [−0.383, −0.089]
OP4 sensitivity ≥ 0.80	0.753 → 0.380	−0.373 [−0.469, −0.270]	+0.278 [+0.203, +0.359]	0.744 → 0.197	−0.546 [−0.671, −0.421]
OP5 sensitivity ≥ 0.90	0.869 → 0.588	−0.281 [−0.391, −0.178]	+0.339 [+0.236, +0.425]	0.862 → 0.400	−0.462 [−0.614, −0.310]

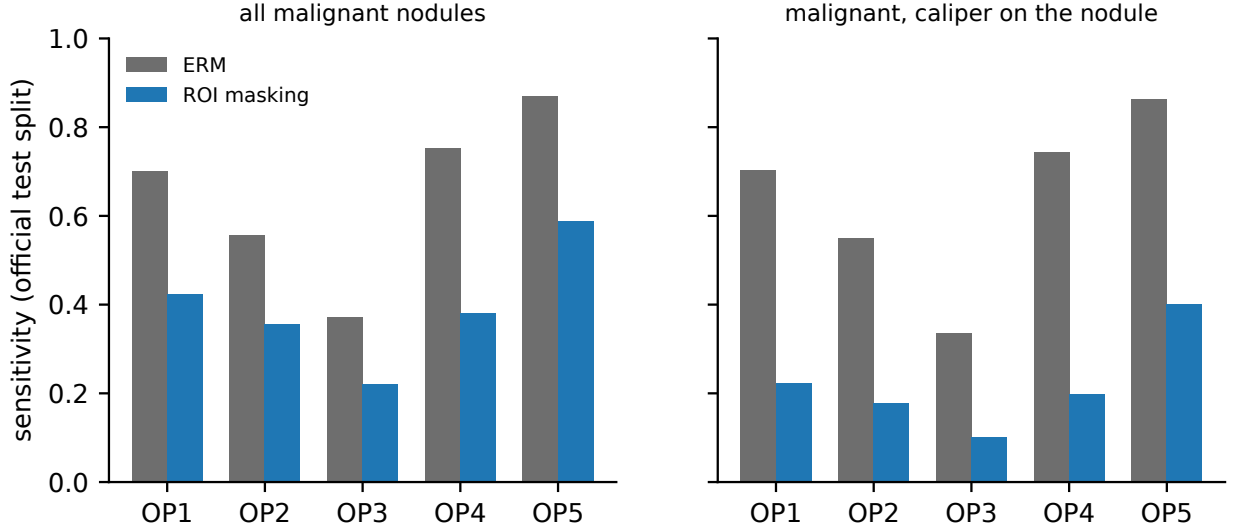


Figure 1: Thyroid, official patient-disjoint test split: sensitivity of ERM and of the masked model at five validation-fixed operating points, for all malignant nodules (left) and for the 78 malignant nodules with a caliper on the nodule (right; post hoc subgroup, exploratory).

Other test sets Table 4. At OP2 masking lowers sensitivity, including for the shortcut-conflicting positives, in the HAM and MSK hold-outs, where it raises specificity; in BCN it lowers specificity; in capsule it raises sensitivity without a significant change for the conflicting erosions. The thyroid pattern — a threshold that no longer catches the positives whose supporting artifact is gone — recurs in two of the four other test sets. Table 5 gives the threshold-free and calibration metrics: in thyroid and BCN masking worsens Brier score and calibration; in HAM, MSK and capsule it improves every metric, matching the AUROC results.

Table 4: Other natural test sets at OP2 (validation specificity ≥ 0.80), mask – ERM, crossed CIs.

Test set	Conflicting positives	Δ sensitivity	Δ specificity	Δ sensitivity, conflicting
Dermoscopy BCN	1832	+0.017 [−0.003, +0.036]	−0.110 [−0.135, −0.079]	+0.019 [−0.004, +0.042]
Dermoscopy HAM	808	−0.062 [−0.110, −0.015]	+0.092 [+0.064, +0.121]	−0.055 [−0.104, −0.007]
Dermoscopy MSK	492	−0.074 [−0.127, −0.026]	+0.083 [+0.064, +0.110]	−0.094 [−0.149, −0.044]
Capsule	38	+0.140 [+0.082, +0.222]	+0.041 [−0.029, +0.107]	+0.022 [−0.041, +0.123]

Table 5: Clinical metrics on the natural test sets (seed means; threshold at maximal balanced accuracy on validation data).

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sensitivity	Specificity
Capsule (held-out frames)	ERM	0.889	0.897	0.130	0.042	0.770	0.844
	mask	0.967	0.971	0.069	0.029	0.856	0.933
	balanced	0.881	0.889	0.136	0.051	0.757	0.838
ISIC, BCN held out	ERM	0.786	0.587	0.238	0.279	0.842	0.521
	mask	0.761	0.564	0.287	0.330	0.842	0.443
	balanced	0.782	0.587	0.229	0.260	0.835	0.509
ISIC, HAM held out	ERM	0.752	0.355	0.161	0.233	0.575	0.789
	mask	0.782	0.396	0.138	0.199	0.569	0.831
	balanced	0.751	0.350	0.154	0.225	0.495	0.845
ISIC, MSK held out	ERM	0.770	0.477	0.171	0.160	0.600	0.790
	mask	0.785	0.497	0.149	0.120	0.587	0.823
	balanced	0.767	0.464	0.175	0.165	0.628	0.759
Thyroid (official split)	ERM	0.735	0.604	0.209	0.088	0.699	0.653
	mask	0.731	0.629	0.215	0.119	0.424	0.856
	balanced	0.714	0.580	0.220	0.115	0.707	0.612

5.3 Replication breadth

Table 9 and Figure 2. The crossover’s interval lies above zero for 17 of 18 frozen encoder-cohort pairs; the exception is DermLIP (Stage 3).

Model scale Table 6. The law holds at every size in every cohort. Larger models do not escape the shortcut; they exploit an in-ROI caliper *more*: ERM’s correlated–reversed gap in the thyroid Trap A grows from 0.537 (ViT-S) to 0.590 (ViT-B) and 0.622 (ViT-L), and in the ovary from 0.320 to 0.564; capsule is flat.

Table 6: Encoder scale (DINOv2 ViT-S/14 21 M, ViT-B/14 86 M, ViT-L/14 304 M parameters): ERM’s shortcut gap in the in-ROI trap and the crossover.

Cohort	DINOv2	ERM clean AUROC (Trap A)	ERM gap corr – rev (Trap A)	Crossover
Thyroid	ViT-S/14	0.629	0.537	+0.247 [+0.205, +0.288]
Thyroid	ViT-B/14	0.608	0.590	+0.239 [+0.196, +0.280]
Thyroid	ViT-L/14	0.601	0.622	+0.263 [+0.218, +0.307]
Ovary	ViT-S/14	0.698	0.320	+0.296 [+0.208, +0.384]
Ovary	ViT-B/14	0.635	0.413	+0.170 [+0.087, +0.252]
Ovary	ViT-L/14	0.608	0.564	+0.207 [+0.127, +0.286]
Capsule	ViT-S/14	0.803	0.363	+0.370 [+0.325, +0.415]
Capsule	ViT-B/14	0.796	0.360	+0.377 [+0.327, +0.426]
Capsule	ViT-L/14	0.820	0.347	+0.417 [+0.369, +0.464]

End-to-end fine-tuning Table 7. The crossover replicates in 3 of 5 fine-tuned networks: dermoscopy hair with ResNet-50 (+0.143 [+0.054, +0.232]), thyroid with ViT-S and capsule with ResNet-50, where masking harms strongly inside the ROI (−0.279 [−0.377, −0.185]) and helps outside (+0.327 [+0.235, +0.416]). The thyroid ResNet-50 points the same way without significance. The fine-tuned ovary network is *not evaluable*: with 15 clusters its ERM clean AUROC is 0.60 (Trap A) and 0.50 (Trap B) — it barely learns the task — so its null says nothing about the law. Fine-tuning does not remove the effect; it is not an artefact of frozen features.

Table 7: End-to-end fine-tuned networks (ImageNet initialisation, 224 px, 8 epochs; folds as bootstrap clusters): effect of masking per trap (crossed CIs).

Fine-tuned network	mask – ERM, Trap A	mask – ERM, Trap B	mask+balanced – balanced, Trap A
Dermoscopy hair, ResNet-50	−0.002 [−0.054, +0.049]	+0.141 [+0.071, +0.214]	−0.008 [−0.072, +0.059]
Thyroid, ResNet-50	+0.078 [+0.032, +0.123]	+0.142 [+0.073, +0.211]	+0.085 [+0.035, +0.137]
Thyroid, ViT-S/16	+0.122 [+0.068, +0.170]	+0.244 [+0.163, +0.329]	+0.059 [+0.003, +0.122]
Capsule, ResNet-50	−0.279 [−0.377, −0.185]	+0.327 [+0.235, +0.416]	−0.020 [−0.107, +0.074]
Ovary, ResNet-50 (15 clusters)	+0.152 [+0.070, +0.232]	+0.211 [+0.090, +0.328]	<i>n/a</i>

Untrained models Table 8. Zero-shot scores of vision–language models are already swayed by artifacts (capsule correlated–reversed gap −0.402), and masking changes them according to the law: crossover thyroid +0.103 [+0.064, +0.143], ovary +0.181 [+0.103, +0.261], capsule +0.045 [+0.001, +0.090] (borderline), dermoscopy (DermLIP) +0.040 [+0.021, +0.058] — 4 of 4 positive. The effect is therefore a property of the representation, not of the trained head. Caveat: MedSigLIP’s zero-shot thyroid score is at chance (clean AUROC 0.451), so that row shows artifact effects on an uninformative score.

Table 8: Zero-shot vision–language scores (no training: cosine to a positive minus a negative class prompt) on the trap test environments, unmasked and masked (crossed CIs). The gap shows the untrained score is already swayed by the artifact; the crossover shows that masking changes it according to the location law.

Cohort	Model	Zero-shot clean AUROC (Trap A)	Correlated – reversed gap	mask effect, Trap A	mask effect, Trap B	Crossover
Thyroid	MedSigLIP	0.451	+0.072	+0.017 [−0.011, +0.045]	+0.120 [+0.078, +0.162]	+0.103 [+0.064, +0.143]
Ovary	MedSigLIP	0.623	−0.126	−0.235 [−0.289, −0.180]	−0.054 [−0.135, +0.030]	+0.181 [+0.103, +0.261]
Capsule	MedSigLIP	0.631	−0.402	+0.004 [−0.021, +0.029]	+0.049 [+0.001, +0.098]	+0.045 [+0.001, +0.090]
Dermoscopy hair	DermLIP	0.697	+0.082	−0.079 [−0.114, −0.044]	−0.039 [−0.070, −0.008]	+0.040 [+0.021, +0.058]

New patients ISIC 2020 met every condition of the registered gate: 32,997 images of 2,056 patients, patient-disjoint folds; smallest cell 53 hair-free melanomas (gate 50). The frozen hair segmenter reaches a hair-free balanced accuracy of 0.93 and pixel IoU 0.60 on held-out ISIC 2019 images. Masking harms with hair on the lesion (−0.064 [−0.128, −0.002]) and helps with hair beside it (+0.186 [+0.118, +0.249]); the crossover is +0.250 [+0.174, +0.327] (registered claim X1, supported). Thyroid cohorts with patient identifiers either lack nodule masks (ThyUS2Path (Hou et al., 2024), whose scanner interface also triggers the caliper detector) or require a data-use agreement (TN-SCUI 2020, Stanford AIMI), and were not used; DDTI (134 images) is too small.

Table 9: Replication breadth of the real-artifact crossover (crossed CIs).

Cohort	Encoder	Training	Crossover [95% CI]
Dermoscopy hair	DINOv2 ViT-S/14	frozen	+0.163 [+0.130, +0.195]
Dermoscopy hair	DINOv2 ViT-B/14	frozen	+0.147 [+0.111, +0.185]
Dermoscopy hair	DINOv2 ViT-L/14	frozen	+0.153 [+0.110, +0.195]
Dermoscopy hair	DermLIP	frozen	+0.014 [−0.037, +0.063]
Thyroid	DINOv2 ViT-S/14	frozen	+0.247 [+0.205, +0.288]
Thyroid	DINOv2 ViT-B/14	frozen	+0.239 [+0.196, +0.280]
Thyroid	DINOv2 ViT-L/14	frozen	+0.263 [+0.218, +0.307]
Thyroid	MedSigLIP-448	frozen	+0.195 [+0.153, +0.238]
Thyroid	ConvNeXt-384	frozen	+0.275 [+0.231, +0.320]
Ovary	DINOv2 ViT-S/14	frozen	+0.296 [+0.208, +0.384]
Ovary	DINOv2 ViT-B/14	frozen	+0.170 [+0.087, +0.252]
Ovary	DINOv2 ViT-L/14	frozen	+0.207 [+0.127, +0.286]
Ovary	MedSigLIP-448	frozen	+0.178 [+0.109, +0.248]
Capsule	DINOv2 ViT-S/14	frozen	+0.370 [+0.325, +0.415]
Capsule	DINOv2 ViT-B/14	frozen	+0.377 [+0.327, +0.426]
Capsule	DINOv2 ViT-L/14	frozen	+0.417 [+0.369, +0.464]
Capsule	MedSigLIP-448	frozen	+0.335 [+0.286, +0.383]
Capsule	ConvNeXt-384	frozen	+0.320 [+0.268, +0.372]
Dermoscopy hair	ResNet-50	fine-tuned	+0.143 [+0.054, +0.232]
Thyroid	ResNet-50	fine-tuned	+0.063 [−0.022, +0.144]
Thyroid	ViT-S	fine-tuned	+0.122 [+0.039, +0.211]
Capsule	ResNet-50	fine-tuned	+0.606 [+0.495, +0.716]
Ovary	ResNet-50 (15 clusters)	fine-tuned	+0.058 [−0.062, +0.179]
Dermoscopy hair, ISIC 2020 (external)	DINOv2 ViT-B/14	frozen, new patients	+0.250 [+0.174, +0.327]

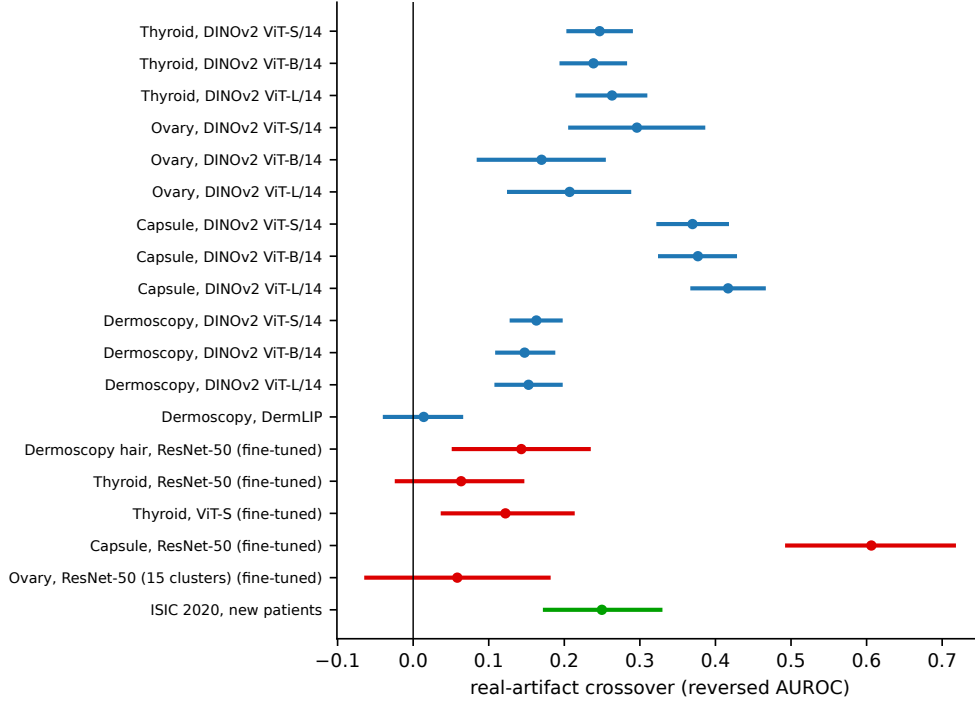


Figure 2: Crossover across encoders and scales (blue), fine-tuned networks (red) and new patients (green); crossed 95% CIs.

Absolute performance Table 10. On the one cohort with a published benchmark on the same patient-disjoint split, the frozen DINOv2 linear probe reaches an AUROC of 0.735, below published fine-tuned CNNs (0.77–0.80). The study is about the *relative* effect of masking on a fixed representation, not about state-of-the-art accuracy, but the gap is a limitation.

Table 10: Absolute performance: the frozen encoder with a linear head against published fine-tuned CNNs on the same patient-disjoint split.

Model (thyroid, official TNCD test split)	AUROC	Source
ResNet-18, cross-entropy (fine-tuned)	0.773	Gong et al. (2022)
DenseNet-121, cross-entropy (fine-tuned)	0.780	Gong et al. (2022)
ConvNeXt-T, cross-entropy (fine-tuned)	0.784	Gong et al. (2022)
ResNet-18, adaptive curriculum (fine-tuned)	0.799	Gong et al. (2022)
DINOv2 ViT-B/14 frozen + linear head, ERM (rebuilt run)	0.735	this work

5.4 The DermLIP exception

With DermLIP, masking harms at both hair locations (Trap A -0.104 $[-0.143, -0.067]$, Trap B -0.090 $[-0.135, -0.046]$). Two measurements explain it. First, masking costs DermLIP disease signal wherever the hair lies: clean-test AUROC changes by -0.066 $[-0.089, -0.043]$ and -0.063 $[-0.094, -0.033]$. Second, the pixel part of its hair shortcut is small at both locations (oracle hair inpainting $+0.033$ $[+0.025, +0.042]$ and $+0.040$ $[+0.015, +0.064]$), whereas group balancing, which also removes correlates of hair, gains $+0.146$ $[+0.122, +0.170]$ and $+0.110$ $[+0.089, +0.132]$: DermLIP carries the hair shortcut mostly through correlates that masking does not remove at either location. In the terms of the theory, masking lowers S at both locations and removes little of $\tilde{A}$ at either, so there is no gap to find. The theory anticipates this from the clean and correlated AUROC alone: predicted crossover $+0.013$, observed $+0.014$.

5.5 The theory, judged by held-out error

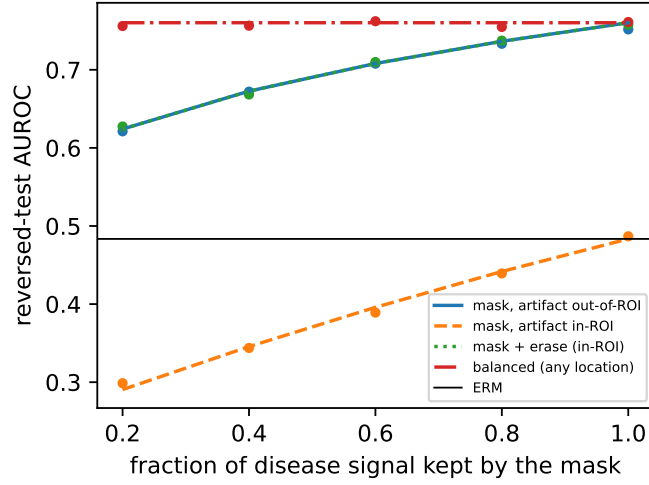


Figure 3: The closed form (lines) against Monte Carlo simulation of its own model (points): the formula is exact up to simulation noise (at most 0.009 AUROC over 29 settings). This checks the algebra, not the theory.

Table 11 and Figure 4. Over 136 primary cells the theory predicts the reversed AUROC — never used in fitting — with mean absolute error 0.052 (Pearson r 0.914; 66% within 0.05), against 0.129 for the symmetric-reversal heuristic and 0.237 for “reversed = clean”. It gets the sign of all 9 real-trap crossovers right (MAE 0.015) and of 3/3 fine-tuned crossovers — networks that violate its assumptions — and the sign of 44 of 50 controlled-sweep gains (heuristic: 38). From the fitted S alone, the sign of the in-ROI masking effect is right in 8 of 9 cells. Its clear failure is the chest-drain cohort of Stage 3: the clean and correlated environments do not reveal a shortcut carried by the treated lung, so the model cannot see it — the boundary its assumption (5) predicts.

The account is simple: a model’s reliance on an artifact is fixed by the ratio of artifact to disease signal it sees; masking lowers the disease signal (less context) and, when the artifact lies inside the ROI, leaves the artifact’s signal intact, so it raises that ratio exactly when the artifact is inside.

Table 11: Held-out prediction of the reversed-test AUROC (never used in fitting): the two-parameter theory fitted to each cell’s clean and correlated AUROC, against two parameter-free heuristics.

Cells	Predictor	n	MAE	Pearson r	within 0.05
Primary cells (ERM, mask)	linear-Gaussian theory	136	0.052	0.914	66%
	heuristic: symmetric reversal	136	0.129	0.743	40%
	heuristic: reversed = clean	136	0.237	0.550	18%
All ERM-head arms	linear-Gaussian theory	598	0.036	0.933	79%
	heuristic: symmetric reversal	598	0.076	0.823	65%
	heuristic: reversed = clean	598	0.151	0.584	40%
Fine-tuned (outside assumptions)	linear-Gaussian theory	12	0.027	0.980	92%

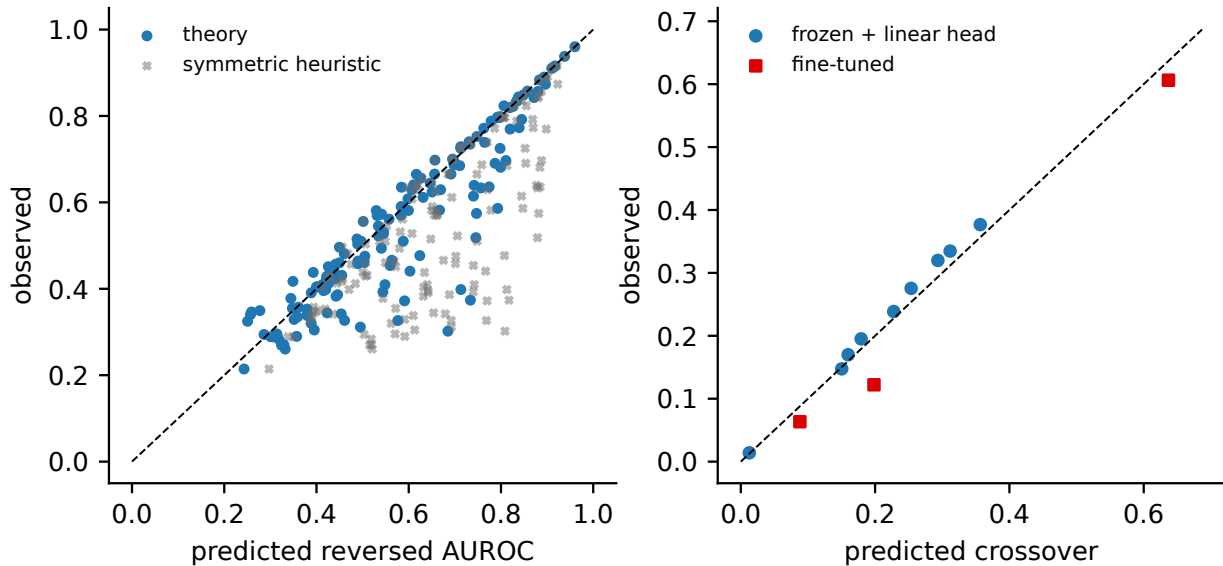


Figure 4: Left: predicted versus observed reversed AUROC (theory, blue; symmetric heuristic, grey). Right: predicted versus observed real-trap crossover (frozen encoders, blue; fine-tuned, red).

Verdict by the registered rule The rule called the theory *quantitatively predictive* only if every crossover sign agreed, both correlations were ≥ 0.7 , and its error beat “reversed = clean”. On the full registered cell set one near-zero chest-radiograph crossover (observed ≈ -0.002) had the wrong sign; the report therefore calls it a *qualitative account that closely matches held-out results* and does not say that it “predicts magnitudes”. The correlation of predicted and observed crossovers ($r = 0.999$) is not used as evidence, because the symmetric heuristic also correlates almost perfectly; the held-out error is the evidence.

5.6 Sensitivity to the interval estimator

On the analyses of this stage (Table 12) the crossed intervals are wider by a median factor of 1.53; 12 of 148 contrasts lose significance and 1 gains it (Table 13). The ones that matter for the text — masking’s gain on the easy thyroid pairs and on all MSK pairs — are therefore not claimed; natural-set intervals are about twice as wide as the per-seed ones, because every seed scores the same test images.

Table 12: The analyses of this stage under the per-seed and the crossed estimator (same predictions; baseline arms only).

Result group	Intervals	Excluded with the estimator	0 only per-seed	Excluded with the estimator	0 only crossed	Median width ra- tio (crossed / per- seed)
capsule/dinol518_scale	9	1		0		1.53
capsule/dinos518_scale	9	0		0		1.38
finetune/capsule	4	0		0		1.10
finetune/isc	4	0		0		1.20
finetune/ovary	2	0		0		1.23
finetune/thyroid	8	1		1		1.17
natural/capsule_dino518_repro	6	0		0		1.16
natural/isc_BCN_dino518_repro	6	0		0		1.72
natural/isc_HAM_dino518_repro	6	0		0		1.95
natural/isc_MSK_dino518_repro	6	2		0		2.01
natural/thyroid_dino518_repro	6	1		0		2.06
ovary/dinol518_scale	9	1		0		1.58
ovary/dinos518_scale	9	1		0		1.56
spec_e13/dinol518_spec_scale	17	2		0		1.44
spec_e13/dinos518_spec_scale	17	1		0		1.61
thyroid/dinol518_scale	9	1		0		1.59
thyroid/dinos518_scale	9	0		0		1.37
zero_shot/boot	12	1		0		2.00
all	148	12		1		1.53

Table 13: Contrasts of this stage whose verdict depends on the estimator.

Result	Contrast	Crossed estimate [95% CI]	Verdict vs per-seed estimator
zero_shot	medsiglip448 / ovary / trapB / zs_mask- zs	-0.054 [-0.135, +0.030]	no longer excludes 0
thyroid/dinol518_scale	trapB / balanced / clean / balanced / erm	+0.036 [-0.002, +0.073]	no longer excludes 0
ovary/dinos518_scale	trapB / balanced / clean / balanced / erm	+0.038 [-0.000, +0.077]	no longer excludes 0
ovary/dinol518_scale	trapB / balanced / clean / balanced / erm	+0.043 [-0.002, +0.087]	no longer excludes 0
capsule/dinol518_scale	trapA / mask / clean / mask / erm	+0.012 [-0.006, +0.031]	no longer excludes 0
spec_e13/dinos518_spec_scale	trapB / mask / clean / all / mask / erm	+0.028 [-0.000, +0.055]	no longer excludes 0
spec_e13/dinol518_spec_scale	trapA / mask / test_rev / HAM / mask / erm	-0.044 [-0.090, +0.002]	no longer excludes 0
spec_e13/dinol518_spec_scale	trapB / mask / clean / all / mask / erm	-0.025 [-0.055, +0.003]	no longer excludes 0
natural/thyroid_dino518_repro	easy / mask-erm	+0.042 [-0.011, +0.097]	no longer excludes 0
natural/isc_MSK_dino518_repro	all / mask-erm	+0.016 [-0.007, +0.038]	no longer excludes 0
natural/isc_MSK_dino518_repro	all / balanced-mask	-0.018 [-0.041, +0.005]	no longer excludes 0
finetune/thyroid/resnet50	trapB / mask_balanced / balanced	+0.069 [-0.004, +0.143]	no longer excludes 0
finetune/thyroid/	trapA / mask_balanced / balanced	+0.059 [+0.003, +0.122]	now excludes 0
vit_small_patch16_224.augreg_in21k_ft_in1k			

6 What failed or changed

- **Natural test sets:** N3 holds in three of five; HAM goes the other way and capsule is inconclusive.
- **The thyroid subgroup** is a post hoc subgroup; its test was registered only after its AUROC result was known.
- **Fine-tuning:** thyroid ResNet-50 is not significant; the ovary network is not evaluable.
- **DermLIP:** no crossover; explained, and anticipated by the theory, but still an exception.
- **The theory** is a qualitative account by its registered rule and fails for correlate-carried shortcuts (chest drains); the chest-radiograph cells could not be re-estimated and are excluded from its current error figures.
- **Absolute performance** is below task-specific fine-tuned models on thyroid.

- **External validation is dermoscopy only;** no ultrasound or capsule cohort passed the gate.

7 Anticipated questions

Why does AUROC barely change on natural data while sensitivity drops by a quarter? AUROC averages over all positive–negative pairs, and most pairs do not involve the shortcut. A fixed threshold, however, is set on validation data where the caliper helps; after masking the model’s scores for malignant nodules without a supporting caliper fall below it. The hard-pair AUROC and the subgroup sensitivity isolate exactly those cases.

Why does masking help on the hard HAM pairs? This is not explained, and it is reported as a failure of N3 for that test set. A possible reading — that for HAM-style images the model trained on BCN and MSK relied more on hair beside the lesion than on hair on it, so that masking removed more shortcut than it kept — has not been tested.

Why do larger models exploit the caliper more? A plausible reading, not tested directly, is that they represent small, high-contrast structures such as caliper crosses more distinctly (a larger A in the theory), so the linear head finds the shortcut more easily. What is established is only that scale is not a defence.

Why include zero-shot scores whose thyroid AUROC is at chance? Only to show that the location effect lives in the representation: even without any head trained on the trap, masking changes the artifact’s influence differently inside and outside the ROI.

Why is ISIC 2020 external if it is also ISIC? It is a different acquisition campaign with different patients, patient identifiers and no image overlap; the hair masks and lesion masks were produced by models frozen before any ISIC 2020 image was seen. It is external for patients, not for modality.

Is a two-parameter model with two inputs per cell not trivially accurate? It has two inputs and predicts a third quantity it never saw; the heuristics use the same two inputs and are two to four times worse. It also predicts crossovers, which combine four cells, and signs of in-ROI effects.

Why keep the subgroup result if it is post hoc? It is the clinically meaningful case — a malignant nodule the sonographer measured — and the test in it was registered before it was computed. It is labelled exploratory everywhere.

8 Conclusion

Supported. On unaltered data masking harms the shortcut-conflicting cases in three of five test sets and lowers thyroid sensitivity at every validation-fixed operating point (registered decision rule met). The crossover replicates across encoder families and sizes, in three of the four evaluable fine-tuned networks, in untrained vision–language scores and on new patients. A two-parameter theory matches held-out reversed AUROC far better than heuristics, explains every regime observed so far, and anticipates the DermLIP exception. **Qualified.** The 78-nodule subgroup is exploratory confirmation; the theory is a qualitative account by its own rule. **Not supported.** Harm on natural HAM and capsule test sets; a crossover for DermLIP and for the thyroid ResNet-50.

9 Questions for the supervisor

1. Is a sensitivity loss on one patient-disjoint split (thyroid) enough for a clinical claim, or should the thesis lead with the AUROC on shortcut-conflicting cases?
2. Should the theory be a main contribution (with its held-out error as evidence) or a supporting section?
3. Is ISIC 2020 enough external validation, given that it is the same modality and artifact as the main dermoscopy analysis?
4. Should the below-benchmark absolute performance of frozen linear probes be addressed with fine-tuned models in the main text, or is the fine-tuned replication sufficient?

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